

Assignment 6

September 26, 2026

1 Question 1

two masses, each of mass m , are connected by a spring with spring constant k .

(a)

since the two masses are identical, the centre of mass is in the middle of the spring. if one mass moves a distance x away from the centre of mass, the other mass moves a distance x in the opposite direction.

total change in the length of the spring is

$$x + x = 2x$$

from the hint, if the length of a spring is halved, its effective spring constant doubles. so, each half of the spring has spring constant

$$k_{\text{half}} = 2k$$

for one mass, the restoring force is $F = -2kx$
using Newton's second law,

$$F = m \frac{d^2x}{dt^2}$$

$$m \frac{d^2x}{dt^2} = -2kx$$

rearranging,

$$\frac{d^2x}{dt^2} + \frac{2k}{m}x = 0$$

the standard equation for simple harmonic motion is

$$\frac{d^2x}{dt^2} + \omega^2x = 0$$

comparing the two equations gives

$$\omega^2 = \frac{2k}{m}$$

therefore

$$\omega = \sqrt{\frac{2k}{m}}$$

since $\omega = 2\pi f$, we have

$$2\pi f = \sqrt{\frac{2k}{m}}$$

therefore, the resonant frequency is

$$f = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$$

(b)

we are given $m = 1$ AMU and $f = 125$ THz. Converting the mass into SI units,

$$m = 1.66054 \times 10^{-27} \text{ kg}$$

$$f = 125 \times 10^{12} \text{ Hz} = 1.25 \times 10^{14} \text{ Hz}$$

from part (a),

$$f = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$$

multiplying both sides by 2π

$$2\pi f = \sqrt{\frac{2k}{m}}$$

squaring both sides,

$$4\pi^2 f^2 = \frac{2k}{m}$$

$$k = 2\pi^2 f^2 m$$

substituting the values,

$$k = 2\pi^2 (1.25 \times 10^{14})^2 (1.66054 \times 10^{-27})$$

$$k \approx 510.6 \text{ N/m}$$

hence,

$$k \approx 5.11 \times 10^2 \text{ N/m or } k \approx 511 \text{ N/m}$$

2 Question 2

we are given:

- $r = 0.05 \text{ kg/s}$
- $x_0 = 10 \text{ cm} = 0.1 \text{ m}$
- $v_0 = 0$
- $k = 5 \text{ N/m}$
- $m = 1 \text{ kg}$

(a)

the equation for a damped mass-spring system is

$$m \frac{d^2 x}{dt^2} + r \frac{dx}{dt} + kx = 0$$

substituting the given values,

$$\frac{d^2 x}{dt^2} + 0.05 \frac{dx}{dt} + 5x = 0$$

the standard form is

$$x'' + 2\gamma x' + \omega_0^2 x = 0$$

comparing coefficients,

$$2\gamma = 0.05$$

$$\gamma = 0.025 \text{ s}^{-1}$$

the undamped angular frequency is

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{5}{1}} = \sqrt{5} \approx 2.236 \text{ rad/s}$$

the damped angular frequency is

$$\omega_d = \sqrt{\omega_0^2 - \gamma^2}$$

$$\omega_d = \sqrt{5 - (0.025)^2}$$

$$\omega_d \approx 2.23593 \text{ rad/s}$$

for an underdamped oscillator, the general solution is

$$x(t) = e^{-\gamma t} [A \cos(\omega_d t) + B \sin(\omega_d t)]$$

therefore,

$$x(t) = e^{-0.025t} [A \cos(2.23593t) + B \sin(2.23593t)]$$

using the initial displacement,

$$x(0) = 0.10$$

at $t = 0$,

$$0.1 = A \cos(0) + B \sin(0)$$

since $\cos(0) = 1$ and $\sin(0) = 0$, we get

$$A = 0.10$$

therefore,

$$x(t) = e^{-0.025t} [0.10 \cos(2.23593t) + B \sin(2.23593t)]$$

differentiating to apply the initial velocity condition,

$$v(t) = -\gamma e^{-\gamma t} [A \cos(\omega_d t) + B \sin(\omega_d t)] + e^{-\gamma t} [-A\omega_d \sin(\omega_d t) + B\omega_d \cos(\omega_d t)]$$

at $t = 0$,

$$v(0) = -\gamma A + B\omega_d$$

since $v(0) = 0$,

$$0 = -\gamma A + B\omega_d$$

$$B = \frac{\gamma A}{\omega_d}$$

substituting the values,

$$B = \frac{(0.025)(0.10)}{2.23593}$$

$$B \approx 0.001119$$

therefore, the position as a function of time is

$$x(t) = e^{-0.025t} [0.10 \cos(2.23593t) + 0.001119 \sin(2.23593t)] \text{ m}$$

(b)

initially, the mass is displaced by $x_0 = 0.10$ m and has zero velocity.
therefore, the initial energy is entirely spring potential energy

$$E_0 = \frac{1}{2} k x_0^2.$$

substituting,

$$E_0 = \frac{1}{2} (5) (0.10)^2$$

thus,

$$E_0 = \frac{1}{2} (5) (0.01)$$

and

$$E_0 = 0.025 \text{ J.}$$

the damped angular frequency is

$$\omega_d = 2.23593 \text{ rad/s}$$

therefore, the oscillation period is

$$T = \frac{2\pi}{\omega_d}.$$

substituting,

$$T = \frac{2\pi}{2.23593} \approx 2.8101 \text{ s}$$

after 10 periods,

$$t = 10T$$

so

$$t = 10(2.8101) \approx 28.101 \text{ s}$$

the amplitude decays according to

$$A(t) = A_0 e^{-\gamma t}$$

since the energy is proportional to the square of the amplitude,

$$E(t) = E_0 e^{-2\gamma t}$$

therefore,

$$E(t) = 0.025 e^{-2(0.025)(28.101)}$$

calculating the exponent,

$$2(0.025)(28.101) = 1.40505$$

thus,

$$E(t) = 0.025 e^{-1.40505}.$$

this gives

$$E(t) \approx 0.006134 \text{ J}$$

therefore, the energy lost is

$$\Delta E = E_0 - E(t)$$

hence,

$$\Delta E = 0.025 - 0.006134$$

$$\boxed{\Delta E \approx 0.01887 \text{ J}}$$

3 Question 3

the pipe is closed at one end and open at the other. at the closed end there is a displacement node, while at the open end there is a displacement antinode. for the fundamental mode, the pipe contains one quarter of a wavelength:

$$L = \frac{\lambda_1}{4}$$

$$\lambda_1 = 4L$$

using $f = \frac{c}{\lambda}$,

$$f_1 = \frac{c}{4L}$$

for higher modes, only odd harmonics are allowed. therefore,

$$n = 1, 3, 5, 7, \dots$$

$$L = \frac{n\lambda_n}{4}$$

$$4L = n\lambda_n,$$

$$\lambda_n = \frac{4L}{n}$$

we have,

$$f_n = \frac{c}{\lambda_n}$$

substituting for λ_n ,

$$f_n = \frac{c}{4L/n}$$

hence,

$f_n = \frac{nc}{4L}, \quad n = 1, 3, 5, 7, \dots$
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4 Question 4

we have,

- speed of dolphin, $v_s = 40 \text{ km/h} = 40 \left(\frac{1000}{3600} \right) = 11.111 \text{ m/s}$
- frequency produced by sound, $f_s = 1000 \text{ Hz}$

the Doppler-shifted frequency is

$$f_{\text{heard}} = f_s \frac{v}{v - v_s},$$

where v is the speed of sound in the medium.

(a)

the speed of sound in water is approximately $v = 1500$ m/s. therefore,

$$f_{\text{heard}} = 1000 \left(\frac{1500}{1500 - 11.11} \right)$$
$$f_{\text{heard}} \approx 1007.5 \text{ Hz}$$

therefore,

$$\boxed{f_{\text{heard}} \approx 1008 \text{ Hz}}$$

(b)

taking the speed of sound in air to be approximately $v = 343$ m/s and substituting in $f_{\text{heard}} = f_s \frac{v}{v - v_s}$,

$$f_{\text{heard}} = 1000 \left(\frac{343}{343 - 11.11} \right)$$
$$f_{\text{heard}} \approx 1033.5 \text{ Hz}$$

therefore,

$$\boxed{f_{\text{heard}} \approx 1034 \text{ Hz}}$$